

ALBEN ANTAPPAN

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EDUCATION

Indiana University Bloomington, United States - *Master of Science, Data Science* - (GPA: 3.76)

Aug 2024 — May 2026

University of Mumbai, India - *Bachelor of Engineering, Information Technology* - (GPA: 8.54/10)

Aug 2017 — Jun 2021

EXPERIENCE

School of Medicine, Indiana University

Nov 2025 — Present

Graduate Research Assistant, Bloomington, United States

- Deployed a clinical NLP pipeline on H100 GPU partitions (HPC) using GPT and LLaMA transformer models to extract structured variables from unstructured clinical reports, evaluating output accuracy against expert-labeled ground truth.
- Built Python signal denoising and feature extraction modules for multi-modal neuroimaging data (MRI, fMRI, DTI) across 300+ subjects, producing analysis-ready datasets for cohort-level statistical modeling.
- Automated neuroimaging preprocessing workflows (fMRIPrep, QSIprep/QSIREcon) on an HPC cluster, standardizing raw DICOM data to BIDS format to ensure consistent, high-quality inputs across the full subject cohort.

Kelley School of Business, Indiana University

May 2025 — Aug 2025

Graduate Research Assistant, Bloomington, United States

- Engineered Python ETL pipelines to ingest, clean, and transform large-scale FCC broadband and U.S. Census data across 10,000+ school districts, resolving schema inconsistencies and building the district-level data model.
- Built R-based data wrangling pipelines to harmonize 5 quarters of nurse engagement survey data across 70+ hospital units, producing statistical correlation matrix visualizations across 13 dimensions for a healthcare operations study.

Capgemini

Jul 2021 — Jul 2024

Senior Analyst / Software Engineer, Mumbai, India

- Engineered SQL and Python-based ETL pipelines to automate financial reporting across 7 business units and 25+ global regions, replacing error-prone Excel workflows and improving reporting accuracy by 15%, enabling leadership to make faster, data-driven decisions at scale.
- Developed interactive Power BI dashboards tracking global P&L and financial KPIs for senior leadership, backed by Flask APIs and optimized SQL pipelines, cutting ad-hoc reporting requests and delivering real-time business intelligence.
- Led end-to-end POC for migrating legacy ETL workflows and analytics dashboards to Microsoft Fabric, redesigning the data transformation layer and validating the full system architecture to reduce query latency and improve scalability.
- Tuned ARIMA model parameters and engineered input features for time-series forecasting simulations, validating model performance against holdout data and optimizing execution pipeline to reduce runtime by 25%.

SKILLS

Core Skills	: Python, SQL, NoSQL, R, Statistical Modeling, Machine Learning, Predictive Modeling, Feature Engineering, Hypothesis Testing, Model Deployment, Model Validation
Frameworks & Libraries	: Pandas, NumPy, Scikit-Learn, XGBoost, PySpark, PyTorch, FastAPI, dbt, Streamlit, Matplotlib, Seaborn, Plotly
Generative AI & LLM	: LangChain, LangSmith, Arize, HuggingFace, Transformers, Embeddings, FAISS, RAG Pipelines
Cloud & Platforms	: AWS S3, AWS Lambda, AWS SageMaker, BigQuery, Snowflake, Microsoft Fabric, Microsoft Azure
DevOps & Tools	: Docker, Git, GitHub Actions, CI/CD, Bash Scripting, SLURM
Visualization	: Power BI, Power Query, Tableau, Looker, Grafana, DAX
Certifications	: GCP Certified Professional ML Engineer, Microsoft Power BI Data Analyst, IBM Data Science

PROJECTS

FinSight AI: SEC Filing Analysis Platform, [Link](#)

- Built a production RAG pipeline using FAISS vector search, transformer-based BGE embeddings, and MMR retrieval over SEC 10-K/10-Q filings, serving Llama 3.1 8B to generate structured financial summaries and stream multi-turn conversational Q&A responses.
- Engineered live market data enrichment via yfinance injection into LLM context, implemented per-session conversation memory for stateful multi-turn interactions, and integrated LangSmith for end-to-end observability, enabling production monitoring of retrieval quality, latency, and model outputs across the full pipeline.

U.S. Traffic Safety Intelligence Pipeline, [Link](#)

- Architected an event-driven pipeline on AWS where CSV uploads to S3 automatically trigger a Lambda function that invokes EC2 via SSM, running distributed PySpark jobs to clean, validate, and feature-engineer 500,000+ US accident records.
- Configured SageMaker Autopilot to train, deploy, and serve accident severity prediction models via a live inference endpoint, with SNS alerts on pipeline completion and Power BI connected for downstream business intelligence reporting.

Public Transit On-Time Performance Dashboard, [Link](#)

- Processed and modeled 1.9 million+ GPS and APC transit records entirely within Power BI using Power Query, building a star-schema data model with fact and dimension tables to support high-granularity on-time performance analysis.
- Developed custom DAX measures to define OTP KPIs and built a multi-page interactive dashboard with ArcGIS geospatial mapping, peak vs. non-peak comparative analysis, and route-level breakdowns, refined through direct stakeholder feedback from the transit agency.

PUBLICATIONS

Lie Detection Using Facial Analysis, Electrodermal Activity, Pulse and Temperature, JETIR

Research on Lie Detector Using Facial Analysis and Heartbeat Sensing, IJARESM

Alexa Based Smart Home Monitoring System, IEJRD